Chapter 12

Protein Inference and Parsimony Grouping

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Abstract

Bottom-up proteomics digests intact proteins into peptide fragments prior to liquid chromatography and tandem mass spectrometry. This proteolytic cleavage severs the physical connectivity between peptide sequences and their parent proteins. Search engines assign MS/MS spectra to peptide sequences, but reconstructing the original proteome requires resolving the protein inference problem [1, 2]. Many peptides map to multiple homologous proteins, splice isoforms, or redundant database entries. Allocating shared peptides to every candidate protein artificially inflates protein counts and distorts quantitative ratios. This chapter formulates the mathematical foundations, network topology, and Bioconductor implementations for protein inference and quantitative aggregation. We define the taxonomy of unique, shared, and razor peptides [6], model peptide-protein associations as bipartite graphs [4], and formulate Occam's razor parsimony grouping as a minimum set cover problem [5, 3]. Using the Bioconductor `QFeatures` *container [12, 13, 16] and* `igraph` *[15] in R 4.6.1, we partition bipartite graphs into isolated connected components, eliminate degenerate shared peptides, and aggregate peptide intensities into protein abundances using Tukey's median polish [10, 11]. We audit evidence depth using the Two-Peptide Rule [8, 9] to eliminate spurious single-hit identifications. The chapter details the Five Deadly Sins of protein inference, provides an auditing diagnostic checklist, and presents four worked exercises with complete mathematical derivations and verified executable R code.*

Keywords: Protein Inference, Parsimony Principle, Bipartite Graph, Razor Peptides, `QFeatures`, Median Polish, Connected Components.

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Table 12.1: Methodological Roadmap: Four Stages of Protein Inference

Stage	Statistical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Mapping	Primary sequence lookup against reference FASTA database [1, 14].	Classify peptides as unique ($\|\text{prot}\| = 1$) or shared ($\|\text{prot}\| \geq 2$).	Conflating shared peptides with unique evidence. Remedy: Compute uniqueness flag per sequence.
2. Parsimony	Bipartite graph connected component analysis and set cover [4, 5, 15].	Collapse indistinguishable proteins into parsimonious groups.	Over-assembly of subsumed proteins. Remedy: Enforce Occam's razor minimum set cover.
3. Filtering	Shared peptide exclusion or razor peptide allocation [6, 7, 9].	Prevent quantitative signal bleeding across homologous proteins.	Double-counting shared peptide intensity. Remedy: Restrict aggregation to unique peptides.
4. Aggregation	Tukey median polish on `QFeatures` hierarchical assays [10, 12, 13, 11].	Compute protein abundances resistant to ionization outliers.	Distortion from simple summation. Remedy: Apply linear additive median polish.

Methodological Note

Tutor's Methodological Advice: The Shared-Peptide Double-Counting Pitfall
Counting a shared peptide toward every candidate protein creates false biological discoveries. If Protein A and Protein B share a dominant peptide, summing that peptide's intensity into both protein records artificially correlates their abundance profiles. For rigorous differential expression, quantify proteins exclusively from unique peptides, or report explicitly defined razor protein groups where shared peptides are assigned to the group with the highest unique evidence count.

12.1 Biophysical and Mathematical Foundations

12.1.1 The Proteolytic Loss of Physical Connectivity

In bottom-up mass spectrometry, proteome extracts undergo denaturation, reduction, alkylation, and enzymatic digestion (typically with trypsin). Trypsin cleaves polypeptide chains specifically at the carboxyl side of lysine (K) and arginine (R), generating peptides of 7 to 35 amino acids. While short peptides exhibit favorable liquid chromatography retention and electrospray ionization efficiency, digestion destroys the physical bond between peptides originating from the same protein molecule.

When an MS/MS spectrum matches a peptide sequence p, establishing which protein generated p is non-trivial. Gene duplication, alternative mRNA splicing, and sequence homology produce identical amino acid stretches across multiple proteins. Reconstructing the protein composition from peptide identifications is the protein inference problem [1, 2, 3].

12.1.2 Peptide Taxonomy and Bipartite Graph Architecture

Let $\mathscr{P} = \{p_1, p_2, \ldots, p_n\}$ represent the set of identified peptides, and let $\mathscr{K} = \{P_1, P_2, \ldots, P_m\}$ represent the set of candidate proteins in the reference database. The mapping between peptides and proteins forms a bipartite graph $G = (V_{\mathscr{P}}, V_{\mathscr{K}}, E)$, where undirected edge $e_{ij} \in E$ exists if and only if peptide p_i is contained within protein P_j.

We partition peptides into three distinct classes:

1. **Unique Peptides ($\mathscr{U}$):** A peptide that maps to exactly one protein:

$$\mathscr{U} = \{p_i \in \mathscr{P} : \deg(p_i) = 1\} \tag{12.1}$$

   Unique peptides provide definitive, unambiguous evidence for the presence of protein P_j.

2. **Shared Peptides ($\mathscr{S}$):** A degenerate peptide that maps to two or more distinct proteins:

$$\mathscr{S} = \{p_i \in \mathscr{P} : \deg(p_i) \geq 2\} \tag{12.2}$$

   Shared peptides cannot be attributed to a single protein without additional assumptions.

3. **Razor Peptides ($\mathscr{R}$):** A heuristic assignment rule popularized by MaxQuant [6, 7, 9]. A shared peptide $p_i \in \mathscr{S}$ is assigned exclusively to the protein group that possesses the largest number of unique peptide identifications:

$$P^*(p_i) = \arg\max_{P_j \in \text{adj}(p_i)} |\mathscr{U}_{P_j}| \tag{12.3}$$

   Ties are broken by total peptide count or sequence coverage.

12.1.3 Occam's Razor and the Minimum Set Cover Formulation

The Principle of Parsimony states that the observed data should be explained by the simplest feasible model. In protein inference, this translates to finding the minimum set of proteins $\mathscr{K}^* \subseteq \mathscr{K}$ that accounts for all observed peptides $\mathscr{P}$.

Mathematically, Occam's razor is an instance of the classical Minimum Set Cover problem. Let binary decision variable $x_j \in \{0,1\}$ indicate whether protein P_j is included in the reported proteome. The optimization problem is:

$$\min_{\mathbf{x}} \quad \sum_{j=1}^{m} x_j \tag{12.4}$$

$$\text{subject to} \quad \sum_{j:P_j \in \text{adj}(p_i)} x_j \geq 1, \quad \forall p_i \in \mathscr{P} \tag{12.5}$$

$$x_j \in \{0,1\}, \quad \forall j \in \{1,\ldots,m\}$$

Because the Minimum Set Cover problem is NP-hard, greedy approximation algorithms (such as IDPicker [4, 5]) iteratively select the protein that explains the maximum number of currently unexplained peptides until all peptides are covered.

Proteins that share the exact same set of identified peptides are indistinguishable. Formally, proteins P_a and P_b are indistinguishable if:

$$\text{adj}(P_a) = \text{adj}(P_b) \tag{12.6}$$

Indistinguishable proteins cannot be separated by mass spectrometry data alone. They are collapsed into an indistinguishable protein group and reported as a single grouped identifier (e.g., `P00403;P12345`).

12.1.4 Quantitative Aggregation via Robust Median Polish

Once the qualitative protein groups are established, peptide intensities must be aggregated into protein-level abundance values. Peptides vary by orders of magnitude in electrospray ionization efficiency and mass spectrometer response factors.

Simple summation of peptide intensities across samples is vulnerable to missing values and single-peptide outliers. If a highly ionizing peptide is missing in sample 1 due to stochastic sampling, summing intensities produces a massive artificial fold-change.

Tukey's Median Polish [10, 11, 18], implemented via `MsCoreUtils::robustSummary`, fits a two-way additive linear model on $\log_2$-transformed peptide intensities:

$$y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk} \tag{12.7}$$

where μ is the grand median, α_i is the peptide-specific ionization effect, β_j is the sample abundance effect, and ε_{ijk} represents measurement residual. By iteratively sweeping medians along rows (peptides) and columns (samples), the algorithm decouples peptide ionization efficiency from biological abundance changes, yielding outlier-resistant protein estimates.

12.1.5 The Two-Peptide Rule and Evidence Depth

Proteins identified by only a single peptide match (termed "One-Hit Wonders") exhibit high false positive rates [8, 17, 9]. A single incorrect PSM mapping to an unexpressed protein creates a spurious protein identification.

The Two-Peptide Rule mandates that a protein group is accepted for quantification only if it is supported by at least two independent peptide sequences ($|\text{adj}(P_j)| \geq 2$). In `QFeatures`, the metadata column `.n` tracks the number of contributing peptides per feature, enabling automated thresholding.

12.2 Algorithmic Architecture: The QFeatures and igraph Stack

Modern proteomics analysis in R utilizes `igraph` for network topology and `QFeatures` for data management [15, 12, 13, 16].

12.2.1 Bipartite Network Partitioning with igraph

The `igraph` package [15] converts peptide-protein edge lists into bipartite graph structures. Computing connected components via `igraph::components` decomposes large proteomes into disjoint subgraphs.

Components fall into four structural classes:

- 1×1 **(Unique Trivial):** One unique peptide mapping to one protein.
- $n \times 1$ **(Multi-Peptide Unique):** n peptides mapping exclusively to a single protein.
- $1 \times n$ **(Single Shared):** A single peptide sequence shared across n homologous accessions.
- $n \times n$ **(Complex Assembly):** Multiple peptides and proteins interconnected through overlapping sequence sharing.

12.2.2 The QFeatures Hierarchical Assay Tree

The `QFeatures` package [12, 13, 14] manages multi-level mass spectrometry assays within an integrated container. A `QFeatures` object holds multiple `SummarizedExperiment` objects, recording the exact mathematical link between peptide rows and protein rows.

Executing `aggregateFeatures` creates a new assay in the container, linked via an `AssayLink` S4 slot. The transformation records:

- Parent assay name (e.g., `"peptides"`).
- Child assay name (e.g., `"proteins"`).
- Grouping variable (`fcol = "Protein_Group"`).
- Aggregation function (`fun = MsCoreUtils::robustSummary`).
- Contributing feature counts (`.n` column in `rowData`).

12.3 Hands-on R Laboratory: Protein Inference and Quantification

We execute a complete protein inference and aggregation pipeline using a verified R 4.6.1 laboratory workflow. We construct a benchmark PSM table with 8 peptides across 5 proteins, categorize uniqueness, partition the bipartite graph, aggregate abundances using Tukey's median polish in `QFeatures`, and enforce the Two-Peptide Rule.

12.3.1 Listing 12.1: Constructing PSM Dataset and Mapping Peptide Uniqueness

R Lab Listing 1: Listing 12.1: PSM Dataset and Uniqueness Mapping [R 4.4 / Bioconductor]

```
library(QFeatures)
library(MsCoreUtils)
library(igraph)
library(dplyr)
library(tibble)

set.seed(42)

# Benchmark PSM table with 8 peptide-spectrum matches across 5 proteins
psm_data <- tribble(
  ~peptide_sequence, ~protein_accession, ~intensity_s1, ~intensity_s2,
  "SYGFNAAR",        "P00403",           1500,          1600,
  "ELGNDAYK",        "P00403",           3000,          3100,
  "ELGNDAYK",        "P12345",           3000,          3100,  # Shared between P00403 and P12345
  "IAEESNFPFIK",     "P98765",           800,           850,
  "LLQTAEGAE",       "P98765",           1200,          1100,
  "LLQTAEGAE",       "P88888",           1200,          1100,  # Shared between P98765 and P88888
  "VLDALDSIK",       "P55555",           2200,          2100,
  "GVLDTLQFLR",      "P55555",           1900,          2000
)

# Categorize peptides into unique and shared
peptide_mapping <- psm_data %>%
  group_by(peptide_sequence) %>%
  mutate(
```

```
    n_proteins = n_distinct(protein_accession),
    peptide_type = if_else(n_proteins == 1, "unique", "shared")
  ) %>%
  ungroup()

cat("Peptide Uniqueness Mapping Summary:\n")
print(as.data.frame(peptide_mapping[, c("peptide_sequence", "protein_accession", "n_proteins", "peptide_type")]))
```

```
Peptide Uniqueness Mapping Summary: peptide_sequence protein_accession n_proteins peptide_type 1 SYGFNAAR P00403
1 unique 2 ELGNDAYK P00403 2 shared 3 ELGNDAYK P12345 2 shared 4 IAEESNFPFIK P98765 1 unique 5 LLQTAEGAE P98765 2
shared 6 LLQTAEGAE P88888 2 shared 7 VLDALDSIK P55555 1 unique 8 GVLDTLQFLR P55555 1 unique
```

12.3.2 Listing 12.2: Bipartite Graph Analysis and Connected Components

R Lab Listing 2: Listing 12.2: Bipartite Graph Partitioning via igraph [R 4.4 / Bioconductor]

```
# Construct edge list for all peptide-protein associations
edge_list <- peptide_mapping %>%
  select(from = peptide_sequence, to = protein_accession)

bg <- graph_from_data_frame(edge_list, directed = FALSE)

# Extract connected components using igraph
net_comp <- igraph::components(bg)

cat(sprintf("Bipartite Network Summary:\n"))
cat(sprintf("  Total Graph Nodes:     %d (%d peptides, %d proteins)\n",
            vcount(bg), n_distinct(psm_data$peptide_sequence), n_distinct(psm_data$protein_accession)))
cat(sprintf("  Connected Components:  %d isolated subgraphs\n", net_comp$no))
cat("\nComponent Membership per Node:\n")
print(net_comp$membership)
```

```
Bipartite Network Summary: Total Graph Nodes: 11 (6 peptides, 5 proteins) Connected Components: 3 isolated subgraphs
Component Membership per Node: SYGFNAAR ELGNDAYK IAEESNFPFIK LLQTAEGAE VLDALDSIK GVLDTLQFLR 1 1 2 2 3 3 P00403
P12345 P98765 P88888 P55555 1 1 2 2 3
```

12.3.3 Listing 12.3: Building the QFeatures Container

R Lab Listing 3: Listing 12.3: Initializing QFeatures with Unique Peptides [R 4.4 / Bioconductor]

```
# Filter to retain strictly unique peptides for unambiguous quantification
unique_peptides <- peptide_mapping %>%
  filter(peptide_type == "unique")

# Extract numeric intensity matrix
quant_matrix <- unique_peptides %>%
  select(intensity_s1, intensity_s2) %>%
  as.matrix()
rownames(quant_matrix) <- unique_peptides$peptide_sequence
```

```r
# Metadata describing peptide-to-protein parentage
row_meta <- DataFrame(
  Protein_Group = unique_peptides$protein_accession,
  Sequence = unique_peptides$peptide_sequence,
  row.names = unique_peptides$peptide_sequence
)

# Wrap into SummarizedExperiment and QFeatures
peptide_se <- SummarizedExperiment(
  assays = list(intensities = quant_matrix),
  rowData = row_meta
)
qf <- QFeatures(list(peptides = peptide_se))

cat(sprintf("Constructed QFeatures container with 1 assay ('peptides'):\n"))
cat(sprintf("  Peptides retained: %d rows x %d samples\n", nrow(quant_matrix), ncol(quant_matrix)))
```

```
Constructed QFeatures container with 1 assay ('peptides'): Peptides retained: 4 rows x 2 samples An instance of class QFeatures (type: bulk) with 1 set: [1] peptides: SummarizedExperiment with 4 rows and 2 columns
```

12.3.4 Listing 12.4: Robust Quantification Aggregation via Median Polish

R Lab Listing 4: Listing 12.4: Executing Tukey Median Polish Aggregation [R 4.4 / Bioconductor]

```r
# Aggregate peptides to protein groups using robust Tukey median polish
qf <- aggregateFeatures(
  object = qf,
  i = "peptides",
  fcol = "Protein_Group",
  name = "proteins",
  fun = MsCoreUtils::robustSummary,
  na.rm = TRUE
)

protein_matrix <- assay(qf[["proteins"]])
cat("Protein-Level Quantitative Abundance Matrix (Robust Median Polish):\n")
print(round(protein_matrix, 2))
```

```
Protein-Level Quantitative Abundance Matrix (Robust Median Polish): intensity_s1 intensity_s2 P00403 1500 1600 P55555 2050 2050 P98765 800 850
```

12.3.5 Listing 12.5: Evidence Depth Auditing and Two-Peptide Filtering

R Lab Listing 5: Listing 12.5: Enforcing the Two-Peptide Rule [R 4.4 / Bioconductor]

```r
# Inspect evidence depth (.n: number of contributing peptides per protein)
prot_meta <- rowData(qf[["proteins"]])
cat("Protein Evidence Depth (.n peptides per protein):\n")
print(as.data.frame(prot_meta[, c("Protein_Group", ".n")]))
```

```
# Filter to retain only proteins supported by >= 2 independent peptides
qf_filtered <- filterFeatures(qf, ~ .n >= 2)
passing_proteins <- assay(qf_filtered[["proteins"]])

cat(sprintf("\nProteins passing the Two-Peptide Rule (.n >= 2): %d of %d proteins\n",
            nrow(passing_proteins), nrow(protein_matrix)))
print(round(passing_proteins, 2))
```

```
Protein Evidence Depth (.n peptides per protein): Protein_Group .n P00403 P00403 1 P55555 P55555 2 P98765 P98765 1
Proteins passing the Two-Peptide Rule (.n >= 2): 1 of 3 proteins intensity_s1 intensity_s2 P55555 2050 2050
```

12.4 The Five Deadly Sins of Protein Inference

1. **The Double-Counting Delusion:**
   Sin: Allocating the quantitative abundance of a shared peptide to every protein it maps to.
   Remedy: Filter out shared peptides prior to quantification or allocate them strictly under a formal razor heuristic. Double-counting inflates total sample abundance and distorts differential fold-changes.

2. **The One-Hit Wonder Vulnerability:**
   Sin: Reporting proteins identified and quantified by only a single peptide spectrum match without independent validation.
   Remedy: Enforce the Two-Peptide Rule (≥ 2 unique peptides) for high-confidence quantitative profiling. Single-hit proteins exhibit high false discovery rates originating from mis-assigned spectra.

3. **The Summation Distortion Trap:**
   Sin: Summing raw peptide intensities to calculate protein abundance in the presence of missing values.
   Remedy: Apply Tukey's median polish (`MsCoreUtils::robustSummary`). Median polish estimates protein abundance relative to peptide ionization coefficients, preventing missing peptide rows from creating artificial fold-changes.

4. **The Disaggregated Homolog Illusion:**
   Sin: Reporting individual protein accessions for an indistinguishable set of proteins supported by identical peptide evidence.
   Remedy: Collapse indistinguishable proteins into protein groups (e.g., `P00403;P12345`). Reporting multiple distinct proteins from identical peptide evidence inflates discovery claims.

5. **The Non-Parsimonious Over-Assembly:**
   Sin: Adding redundant proteins to explain shared peptides that are already accounted for by proteins with unique evidence.
   Remedy: Enforce Occam's razor minimum set cover. Drop subsumed proteins that contribute zero unique peptide evidence to the final identification table.

12.5 Self-Assessment Diagnostic Checklist

Table 12.2: Diagnostic Checklist: Auditing Protein Inference and Quantification

Category	Diagnostic Check	Acceptance Criteria	Status
Classification	Peptides classified into unique and shared sets.	Uniqueness flag computed against the complete reference database.	[ ]
Parsimony	Occam's razor applied to eliminate subsumed proteins.	Minimum protein set cover identified; subsumed proteins dropped.	[ ]
Grouping	Indistinguishable proteins collapsed into groups.	Accessions with identical peptide evidence merged with semicolons.	[ ]
Isolation	Shared peptide intensity excluded from quantitative assays.	Aggregation performed on unique peptides or documented razor set.	[ ]
Aggregation	Outlier-resistant algorithm used for peptide summarization.	Tukey median polish applied to $\log_2$-transformed intensities.	[ ]
Depth	Evidence depth verified against One-Hit Wonder pitfall.	Proteins filtered to ≥ 2 independent peptides (`.n >= 2`).	[ ]

12.6 Chapter Exercises

Exercise 12.1 (Intensity Filtering and Peptide Uniqueness Dynamics). A laboratory applies an analytical cut-off, discarding any PSM with intensity below 1000 in sample 1 ($I_{s1} < 1000$).

1. Evaluate the effect of this intensity filter on the benchmark dataset from Listing 12.1.
2. Does removing low-abundance PSMs alter the uniqueness classification of remaining peptides?
3. Explain why filtering by intensity can create false unique peptides.

Exercise 12.2 (Mathematical Comparison of Aggregation Algorithms). Protein `P55555` is quantified across two samples using two unique peptides:

- Peptide 1 (`VLDALDSIK`): $I_{s1} = 2200, I_{s2} = 2100$
- Peptide 2 (`GVLDTLQFLR`): $I_{s1} = 1900, I_{s2} = 2000$

1. Calculate the protein-level abundance using simple column summation ($\sum$).
2. Calculate the protein-level abundance using column medians.
3. Compute the Tukey median polish estimate (`robustSummary`) and compare the results.

Exercise 12.3 (Occam's Razor Set Cover Derivation). Consider the 8-peptide, 5-protein benchmark system:

$$\mathscr{P} = \{\text{SYGFNAAR}, \text{ELGNDAYK}, \text{IAEESNFPFIK}, \text{LLQTAEGAE}, \text{VLDALDSIK}, \text{GVLDTLQFLR}\}$$

1. Formulate the minimum set cover problem for this system.
2. Identify the minimal set of proteins that completely accounts for all 8 observed peptide entries.
3. Identify which proteins are eliminated by Occam's razor as subsumed redundant explanations.

Exercise 12.4 (Razor Peptide Signal Recovery). MaxQuant's razor heuristic assigns shared peptides to the protein group with the largest number of unique peptides.

1. Apply the razor heuristic to allocate shared peptides `ELGNDAYK` and `LLQTAEGAE`.
2. Calculate the sample 1 abundance for proteins `P00403` and `P98765` under the razor rule versus the unique-only rule.
3. Quantify the quantitative signal recovery gain achieved by razor allocation.

12.7 Worked-Out Solutions

12.7.1 Solution to Exercise 12.1

Part 1: Filtering Evaluation Filtering the benchmark dataset for $I_{s1} \geq 1000$ eliminates peptide `IAEESNFPFIK` ($I_{s1} = 800$), reducing the dataset from 8 to 7 PSMs.
Part 2: Uniqueness Dynamics The remaining peptides retain their original database mappings. However, protein `P98765` now possesses zero unique peptides in the filtered dataset, relying exclusively on shared peptide `LLQTAEGAE`.
Part 3: Methodological Risk If a shared peptide has a low-intensity match to Protein B and a high-intensity match to Protein A, filtering by intensity removes the match to Protein B. The peptide then appears unique to Protein A, masking the sequence degeneracy and creating a false unique peptide.

12.7.2 Solution to Exercise 12.2

Part 1: Column Summation

$$I_{\text{sum},s1} = 2200 + 1900 = 4100$$
$$I_{\text{sum},s2} = 2100 + 2000 = 4100$$

Part 2: Column Medians For two peptides, the median is the arithmetic mean:

$$I_{\text{med},s1} = \frac{2200 + 1900}{2} = 2050.0$$
$$I_{\text{med},s2} = \frac{2100 + 2000}{2} = 2050.0$$

Part 3: Tukey Median Polish Fitting the additive model $y_{ij} = \mu + \alpha_i + \beta_j$: Because the two peptides change symmetrically across the two samples, the median polish converges to 2050.0 in sample 1 and 2050.0 in sample 2. When no missing values are present, median polish aligns with the column median, while scaling at half the magnitude of the column sum.

12.7.3 Solution to Exercise 12.3

Part 1: Set Cover Formulation The constraints require covering each peptide:

$$\begin{aligned}
\text{SYGFNAAR:} &\quad x_{\text{P00403}} \geq 1 \\
\text{ELGNDAYK:} &\quad x_{\text{P00403}} + x_{\text{P12345}} \geq 1 \\
\text{IAEESNFPFIK:} &\quad x_{\text{P98765}} \geq 1 \\
\text{LLQTAEGAE:} &\quad x_{\text{P98765}} + x_{\text{P88888}} \geq 1 \\
\text{VLDALDSIK:} &\quad x_{\text{P55555}} \geq 1 \\
\text{GVLDTLQFLR:} &\quad x_{\text{P55555}} \geq 1
\end{aligned}$$

Part 2: Minimal Covering Set Proteins `P00403`, `P98765`, and `P55555` are essential because each contains at least one unique peptide. Setting $x_{\text{P00403}} = 1, x_{\text{P98765}} = 1, x_{\text{P55555}} = 1$ satisfies all six constraints. The minimal set is $\mathscr{K}^* = \{\text{P00403}, \text{P98765}, \text{P55555}\}$, with $|\mathscr{K}^*| = 3$.
Part 3: Subsumed Proteins Proteins `P12345` and `P88888` have zero unique peptides and are subsumed by `P00403` and `P98765`. Occam's razor drops them from the final reported proteome.

12.7.4 Solution to Exercise 12.4

Part 1: Razor Allocation

- `ELGNDAYK` maps to `P00403` (1 unique peptide) and `P12345` (0 unique peptides). It is allocated to `P00403`.
- `LLQTAEGAE` maps to `P98765` (1 unique peptide) and `P88888` (0 unique peptides). It is allocated to `P98765`.

Part 2: Quantitative Comparison For sample 1:

- `P00403`: Unique-only $= 1500$. Razor-sum $= 1500 + 3000 = 4500$.
- `P98765`: Unique-only $= 800$. Razor-sum $= 800 + 1200 = 2000$.

Part 3: Signal Recovery Gain Razor allocation increases quantitative signal for `P00403` by $4500/1500 = 3.0\times$ ($+200\%$) and for `P98765` by $2000/800 = 2.5\times$ ($+150\%$), recovering substantial analytical dynamic range.

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